

41) Explain why $A \times B \times C$ and $(A \times B) \times C$ are not the same.

Solution: By Defn. 9, $A \times B \times C = \{(a, b, c) \mid a \in A \wedge b \in B \wedge c \in C\}$

By Defn. 8, $A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$

By Defn. 8, $(A \times B) \times C = \{((a, b), c) \mid (a, b) \in A \times B \wedge c \in C\}$

Each member of $A \times B \times C$ is a ordered 3-tuple while that of $(A \times B) \times C$ is a ordered two-tuple where the first member is another ordered 2-tuple.

$\therefore A \times B \times C \neq (A \times B) \times C \quad \square$

42) Explain why $(A \times B) \times (C \times D)$ and $A \times (B \times C) \times D$ are not the same.

Solution: By Defn. 8, $(A \times B) = \{(a, b) \mid a \in A \wedge b \in B\}$

By Defn. 8, $(C \times D) = \{(c, d) \mid c \in C \wedge d \in D\}$

By Defn. 8, $(A \times B) \times (C \times D) = \{((a, b), (c, d)) \mid (a, b) \in A \times B \wedge (c, d) \in C \times D\}$

By Defn. 8, $B \times C = \{(b, c) \mid b \in B \wedge c \in C\}$

By Defn. 9, $A \times (B \times C) \times D = \{(a, (b, c), d) \mid a \in A \wedge (b, c) \in B \times C \wedge d \in D\}$

$(A \times B) \times (C \times D)$ only contains ordered 2-tuple while each member of $A \times (B \times C) \times D$ is an ordered 3-tuple

$\therefore (A \times B) \times (C \times D) \neq A \times (B \times C) \times D$

43) Prove or disprove that if A and B are sets, then $P(A \times B) = P(A) \times P(B)$.

Solution: Let, $A = \{1, 2\}$ $B = \{3, 4\}$

$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$

$P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$ $P(B) = \{\emptyset, \{3\}, \{4\}, \{3, 4\}\}$

$P(A \times B) = \{\emptyset, \{(1, 3)\}, \{(1, 4)\}, \{(2, 3)\}, \{(2, 4)\}, \{(1, 3), (1, 4)\}, \{(1, 3), (2, 3)\}, \{(1, 3), (2, 4)\}, \{(1, 4), (2, 3)\}, \{(1, 4), (2, 4)\}, \{(2, 3), (2, 4)\}, \{(1, 3), (1, 4), (2, 3)\}, \{(1, 3), (1, 4), (2, 4)\}, \{(1, 3), (2, 3), (2, 4)\}, \{(1, 4), (2, 3), (2, 4)\}, \{(1, 3), (1, 4), (2, 3), (2, 4)\}\}$